

Be the Agent

These are the cards for the experiment **Be the Agent**. Print the sheets, cut along the dashed lines and hand them out:

- **pink** cards belong to the Model: the system card, the tasks and the blank cards it writes on;
- **dark** cards are the role cards, one for each player;
- **grey** cards are the material of the tools, which the Model never sees.

Sheet 7 holds blank cards. Print it several times; a game uses twenty or more.

The instructions are on the course website under **Experiments**.

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System

You are the assistant of Greenfield Farm. You can only act by writing cards. On every turn, write exactly one card that starts with **THINK:**, **TOOL:** or **ANSWER:**.

You can use these tools:

- `search("words")`
finds web pages and returns their titles
- `open("title")`
returns the full text of one web page
- `read_file("name")`
returns a farm file: `fields`, `soil_tests`, `stock`, `weather`
- `calculator("expression")`
calculates, e.g. `calculator("4.5 * 100")`
- `write_file("name", "text")`
saves a text as a new farm file

When you have answered the task, write **ANSWER:** and stop.

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Task 1

How large is the field Long Acre, and what grows on it this year?

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Task 2

How much nitrogen may we still apply to Long Acre this spring? Give the total in kg N for the whole field.

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Task 3

Plan the first nitrogen application on Long Acre: how many kilograms of which fertiliser product do we need, and can we spread it on 10 February? Save the plan as the file plan.

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Model

You decide what happens next. On every turn you write exactly one card: `THINK:` for a note to yourself, `TOOL:` for a tool call, `ANSWER:` when you are done.

You see only the stack of cards in front of you, with the system card at the bottom. Nobody talks to you, and you never look at the other tables.

Wait until the result of your last card is on the stack before you write the next one.

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Harness

You carry every card the Model writes.

card	where it goes
THINK:	straight onto the stack
search, open	Search or Web
read_file, write_file	Files
calculator	Calculator
anything else	write <code>RESULT: error, unknown tool</code> yourself
ANSWER:	the game is over

Bring every `RESULT:` card back and put it on top of the stack. Never explain anything to the Model.

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Observer

You keep the log on the Observer's sheet: every card in the order it was written, with a short note on what happened, such as an error, a detour or a good move.

Count the rounds, the tool calls for each tool, the errors and the cards on the stack at the end.

You may look at everything, but you never help, and you never talk to the Model.

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ROLE CARD

Search

You receive `search("words")`.

Compare the words with the keywords in your search index. Answer `RESULT:` followed by the titles of all pages that contain at least one of the words, one title per line.

If no page matches: `RESULT: no pages found`

You know only titles and keywords; you have never seen the pages themselves.

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ROLE CARD

Web

You receive `open("title")`.

If one of your pages has exactly this title, copy its whole text onto a `RESULT: card`, tables included.

Otherwise: `RESULT: error, page not found`

Never correct a title, even if you can guess which page was meant.

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ROLE CARD

Files

You receive `read_file("name")`: copy the whole file onto a `RESULT: card`, or answer `RESULT: error, file not found`.

You receive `write_file("name", "text")`: write the text on a new-file card, keep it with the other files and answer `RESULT: saved`.

A file written this way can be read later like any other.

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ROLE CARD

Calculator

You receive `calculator("expression")`.

Calculate exactly what is written, round to two decimals and answer `RESULT:` with the number.

If you cannot calculate it: `RESULT: error, cannot calculate`

Never calculate anything that was not asked for, even if you can see where the Model is heading.

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Search index

Title	Keywords
Nitrogen requirement of arable crops — teaching guide 2026	nitrogen, requirement, crops, wheat, barley, rape, maize, beet, fertiliser
Fertiliser recommendations 2009 (archive)	nitrogen, fertiliser, wheat, barley, recommendation, archive
Splitting nitrogen in winter wheat	wheat, nitrogen, split, dose, timing, first
When may fertiliser be spread? — teaching rules	when, date, rules, ban, spreading, frozen, snow, fertiliser
Calcium ammonium nitrate (CAN)	CAN, calcium, ammonium, nitrate, product, content, fertiliser
Winter wheat varieties 2026	wheat, varieties, yield, 2026

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fields

Field	Size	Crop 2026
Long Acre	4.5 ha	winter wheat
Mill Field	3.2 ha	oilseed rape
Back Meadow	2.0 ha	grassland

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soil_tests

Field	Sampled	Soil mineral nitrogen (Nmin)
Long Acre	3 Feb 2026	40 kg N/ha
Mill Field	3 Feb 2026	25 kg N/ha

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stock

Product	Nitrogen	In store
Calcium ammonium nitrate (CAN)	27 %	2,000 kg
Urea	46 %	500 kg

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weather

Date	Weather
9 February	frost at night, ground frozen until midday
10 February	frost day and night, ground frozen all day
11 February	mild, ground thawed

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Nitrogen requirement of arable crops — teaching guide 2026

The nitrogen requirement is the amount of nitrogen a crop needs in one season, in kg N per hectare. From it, subtract the soil mineral nitrogen (N_{min}) measured in spring; the rest may be applied as fertiliser.

Crop	Requirement
winter wheat	180 kg N/ha
winter barley	160 kg N/ha
oilseed rape	200 kg N/ha
grain maize	170 kg N/ha
sugar beet	150 kg N/ha

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Fertiliser recommendations 2009 (archive)

This page is no longer updated.

Recommended nitrogen for winter wheat: 210 kg N/ha;
for winter barley: 190 kg N/ha.

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Splitting nitrogen in winter wheat

Winter wheat gets its nitrogen in three doses:

- the first dose, 40 % of the total, at the start of growth in spring;
- the second, 40 %, when the stem starts to grow;
- the third, 20 %, when the ears appear.

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When may fertiliser be spread? — teaching rules

No nitrogen fertiliser may be spread between 1 October and 31 January.

At any time of year, fertiliser must never be spread on ground that is frozen, flooded or covered in snow.

Calcium ammonium nitrate (CAN)

Calcium ammonium nitrate contains 27 % nitrogen: 100 kg of product bring 27 kg of nitrogen to the field.

Winter wheat varieties 2026

Twelve new varieties were tested this year. The highest yields came from varieties with good winter hardiness and resistance to yellow rust.

file: _____

BLANK CARD · MODEL

THINK :

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BLANK CARD · MODEL

TOOL :

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BLANK CARD · TOOL

RESULT :

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ANSWER :

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